

Thomas Tomlinson

 [tjtomlinson](#) |  [thomas-tomlinson](#) |  [thomastomlinson.github.io](#) |  thomas.j.tomlinson@durham.ac.uk |

SUMMARY

I am a third-year PhD student at Durham University studying the stellar dynamics of barred galaxies in cosmological simulations. More broadly, I am interested in galactic archaeology and in using stellar dynamics to test alternative models of dark matter.

EDUCATION

2023 - Sep 2027 (expected) PhD (Astronomy) at **Durham University**
2019 - 2023 Master's Degree (Theoretical Physics) at **Durham University** (First)

PUBLICATIONS

[ADS LIBRARY](#)

First Author

Tomlinson et al. (2026) "Stirring Things Up: Bar-induced substructures in the stellar halo of a cosmological Milky Way analogue", published in MNRAS

TALKS AND CONFERENCE CONTRIBUTIONS

Organiser or LOC member

- **Durham Advanced School on Astrophysical Simulations**
Tutorial Writer & Assistant Sep 2025
- **National Astronomy Meeting (NAM) 2025** Session Organiser & Session Chair Jul 2025

Talks

- **The Computational Galaxy Formation meeting at Ringberg Castle** Invited Talk Jul 2026
- **How Stars Move Around And What They Tell Us About Galaxies' Masses, Formation, and Evolution – From the Milky Way to Giant Elliptical Galaxies** Talk Jul 2026
- **2026 ExGal-Twin Annual Meeting** Talk Apr 2026
- **Observatoire de Paris LIRA Seminar** Talk Mar 2026
- **Auriga Superstars Meeting 2025** Talk Jul 2025
- **Valencia Workshop on the Small-Scale Structure of the Universe and Self-Interacting Dark Matter** Talk Jun 2025
- **The Milky Way and Andromeda: The Odd Couple** Talk Feb 2025
- **2025 ExGal-Twin Annual Meeting** Talk Jan 2025
- **HERA 2024 The Nature and the Dynamics of Structures Observed in Galactic Disks** Talk Sep 2024
- **Auriga Superstars Meeting 2024** Talk Jul 2024
- **The Milky Way Assembly Tale - Plot and characters as of today, and what to expect in future editions** Talk May 2024

- **MPA Cosmology Seminar** Talk Apr 2024
- **Durham-Edinburgh eXtragalactic (DEX) Workshop XX** Short Talk Jan 2024

Posters

- **National Astronomy Meeting (NAM) 2025** Poster Jul 2025
- **Small Galaxies, Cosmic Questions – II** Poster Jul 2024

Attendee

- **ESO Writing and Communicating Your Science** Attendee Aug 2025

RESEARCH SKILLS

Soft skills	Public speaking, science communication, public engagement, data visualisation
Computing skills	Python, C, Agama, AREPO, L ^A T _E X
Hydrodynamical Simulations	Analysis of Auriga simulations, developing a suite of simulations based on the Auriga simulation incorporating SIDM
Observational data	Analysis of data from the Gaia survey, as well as spectroscopic surveys such as APOGEE and LAMOST
Research area	The stellar dynamics of barred galaxies, galactic archaeology, alternative dark matter models

TEACHING EXPERIENCE

Exams marker	Oct 2025 - present
Durham University – Introduction to Astronomy	
Demonstrator	Oct 2024 - Apr 2025
Durham University – Laboratory Skills and Electronics	

OUTREACH AND DEPARTMENT COMMUNITY

RAS Education & Outreach Small Grants Scheme	Co-lead of successful application	Apr 2026
Royal Society Summer Science Exhibition	Exhibit co-organiser	Jan 2026 - Jul 2026
Public seminar talks	Sunday seminar organised by University College Durham	2023 and 2026
CommuniTEA	Departmental social event organiser	Sep 2023 -
Arxiv Coffee	Paper discussion forum	Oct 2023 -
Planetarium	Presenting 'The Northern Sky' planetarium shows to schools and communities	Oct 2024 -
Astronomy Journal Club Co-Chair	Postgraduate journal club	Oct 2024 - Oct 2025